On the Guidance of Flow Matching: review on synthetic data

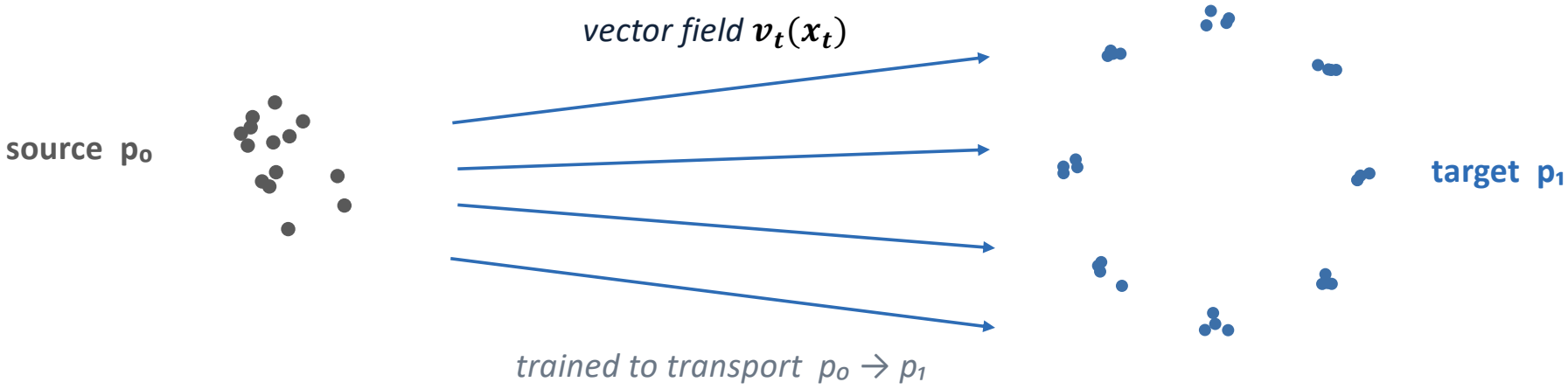
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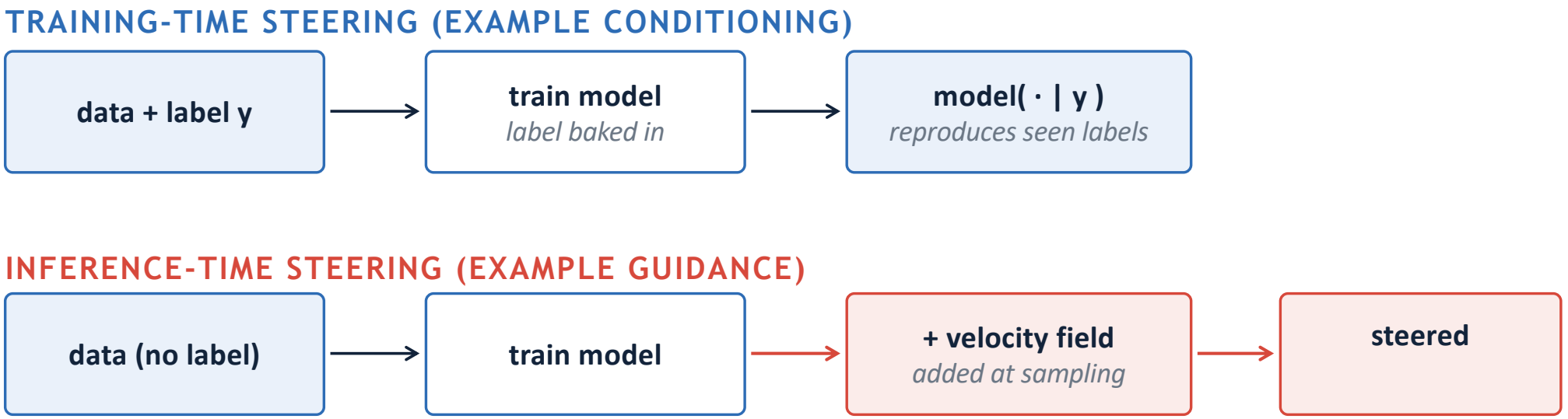
Conditional Flow Matching (CFM) introduction

- $p_0 \rightarrow p_1$: source → target distribution
- $z = (x_0, x_1)$, x_t: sample at time t
- $\mathrm{p}(z) = \pi(x_0, x_1)$, is source–target coupling
- Vector field $v_t(x_t)$ is intractable but $v_{t|z}(x_t|z)$ is tractable in the flow matching loss [2]



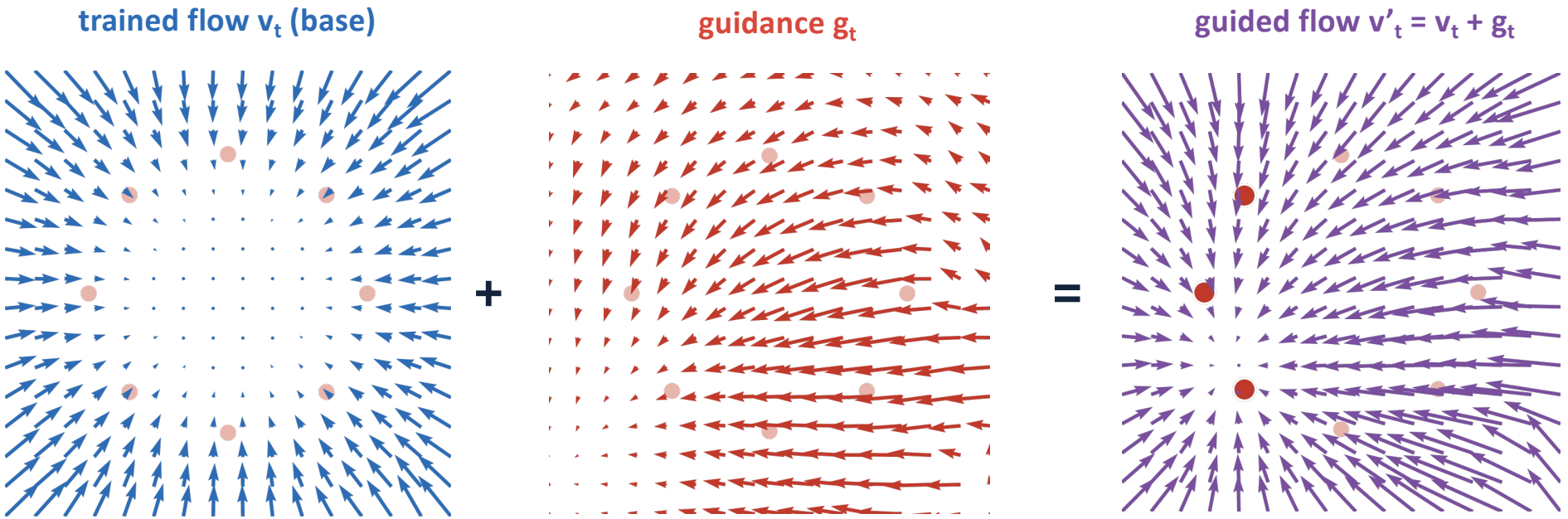
Question: Can we generate parts of a complex target distribution?

Two ways to generate parts of distributions



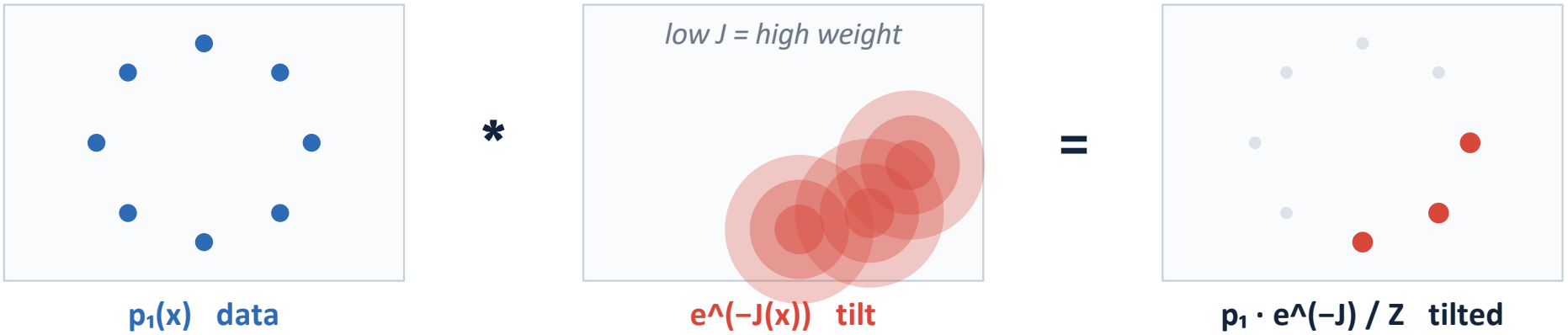
Takeaway: Conditioning needs the label at training.
Guidance steers a frozen model toward an objective you can define after training.

Guidance vector field



$$v'_t(x_t) = g_t(x_t) + v_t(x_t) \quad [3]$$

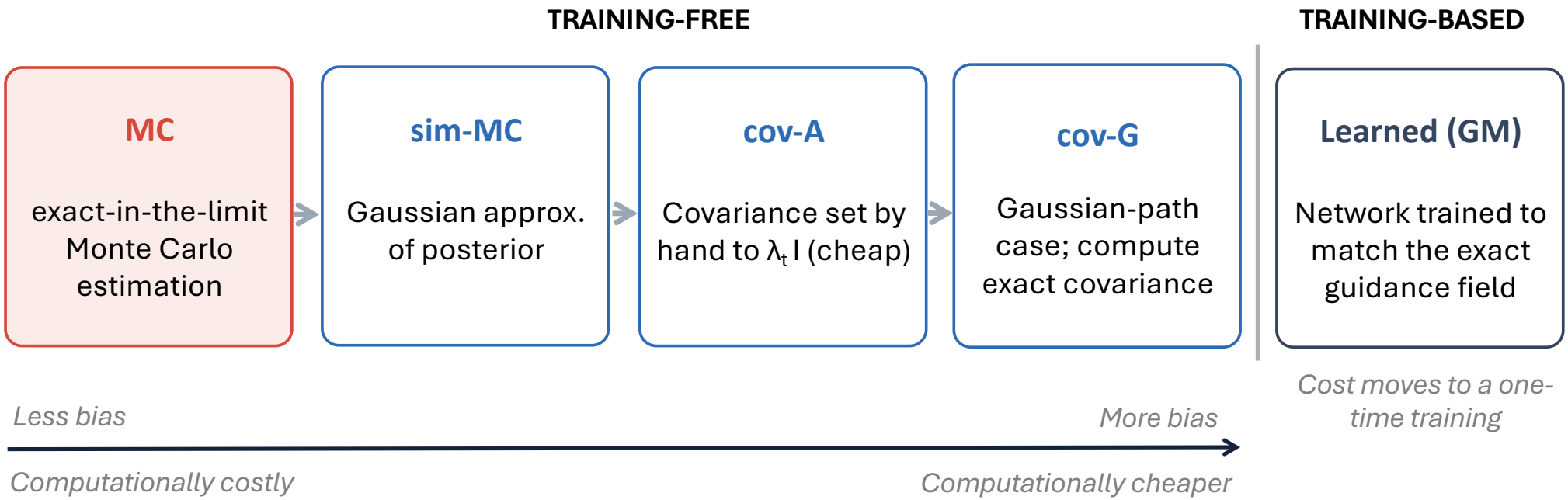
Introduce an energy function J(x)



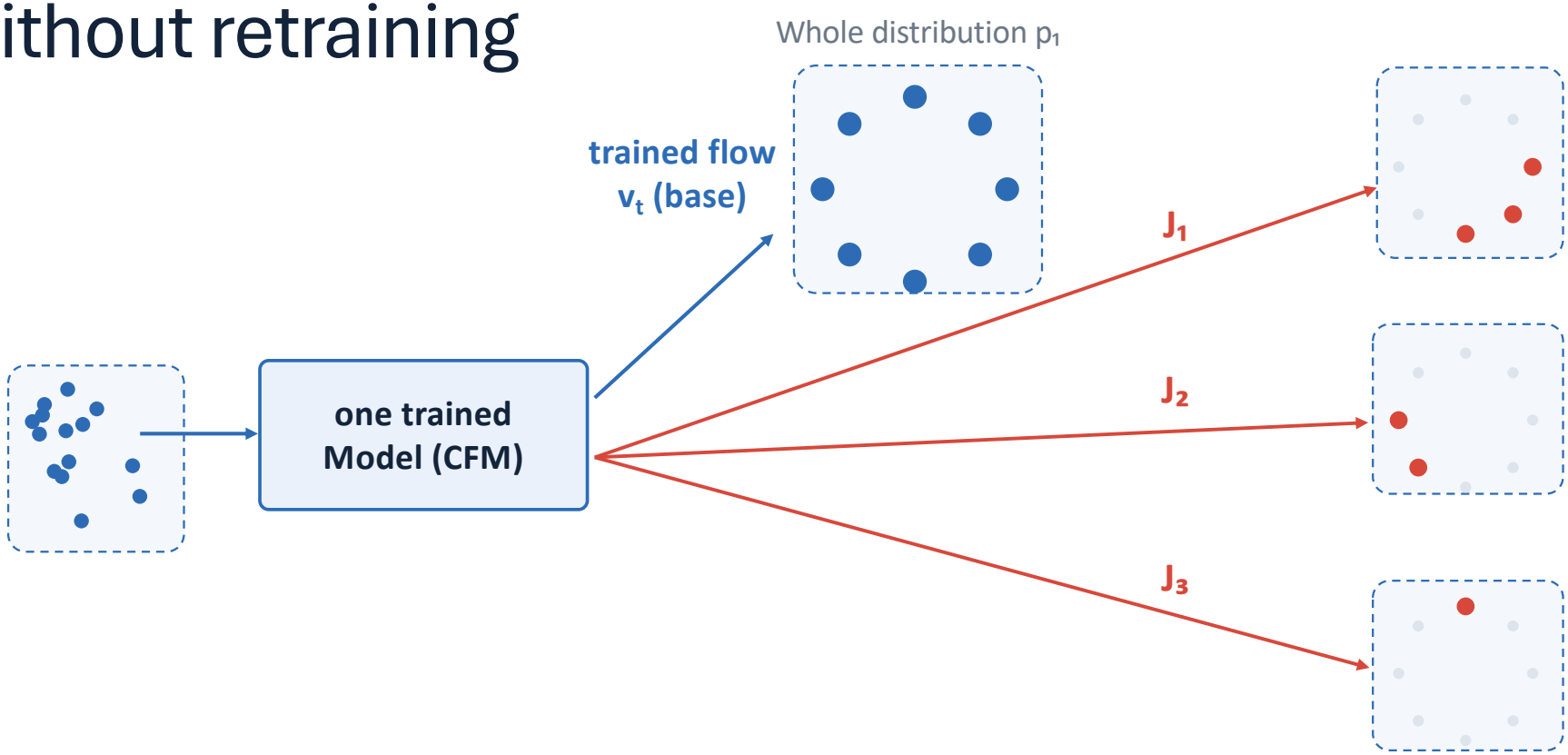
Guided target: $p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$

Takeaway: Guidance reweights the data distribution by e^(−J): the subset is just the low-energy region under J.

Overview of different guidance methods

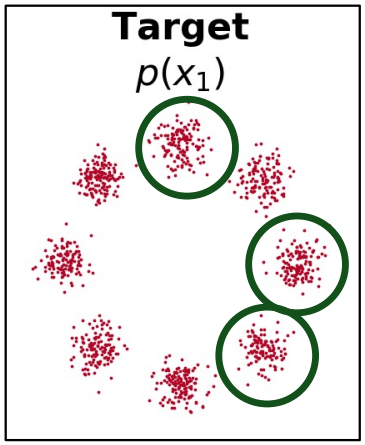


Generating many subsets from the same model without retraining



Takeaway: Train CFM first, then choose an energy function J to generate a subdistribution.

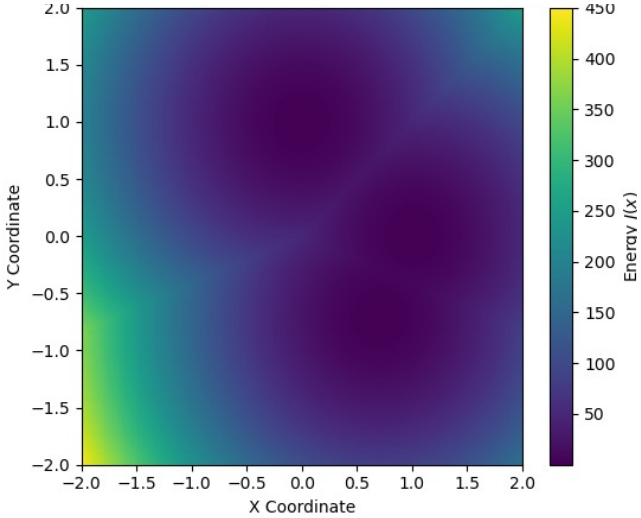
How to construct the energy function



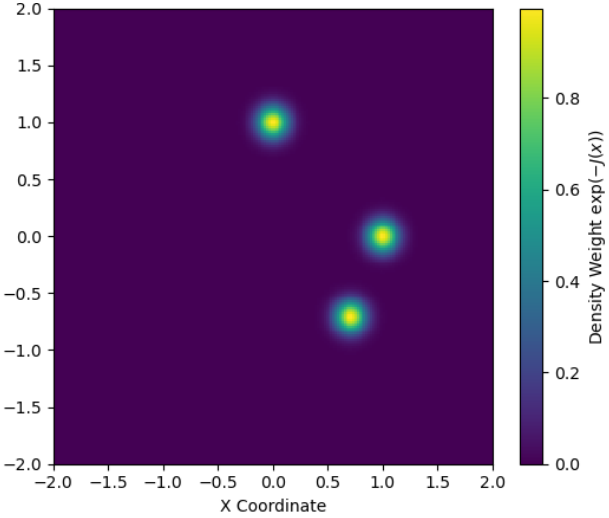
Desired target distribution

8gaussian distribution

Energy function



Exponentially weighted energy function



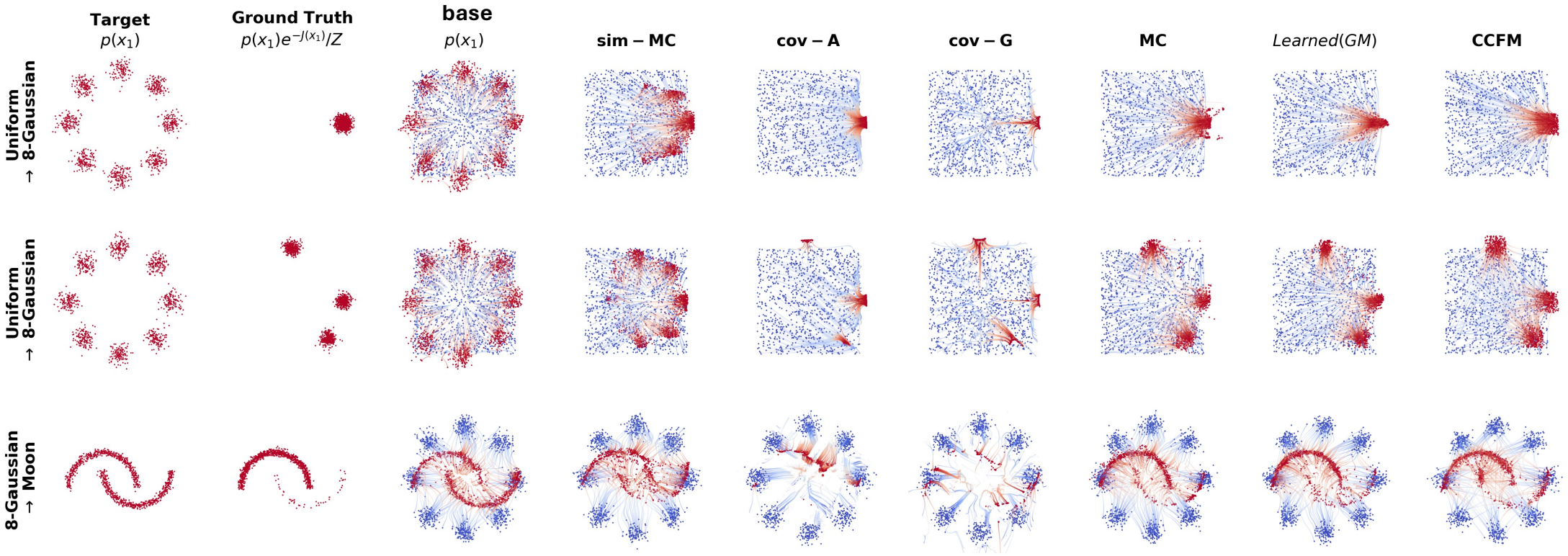
Generated distribution: $p'(x) = \frac{1}{Z}p(x)e^{-J(x)}$

One possible energy function: $J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_3|_2^2}{2\sigma^2}$

σ = energy width

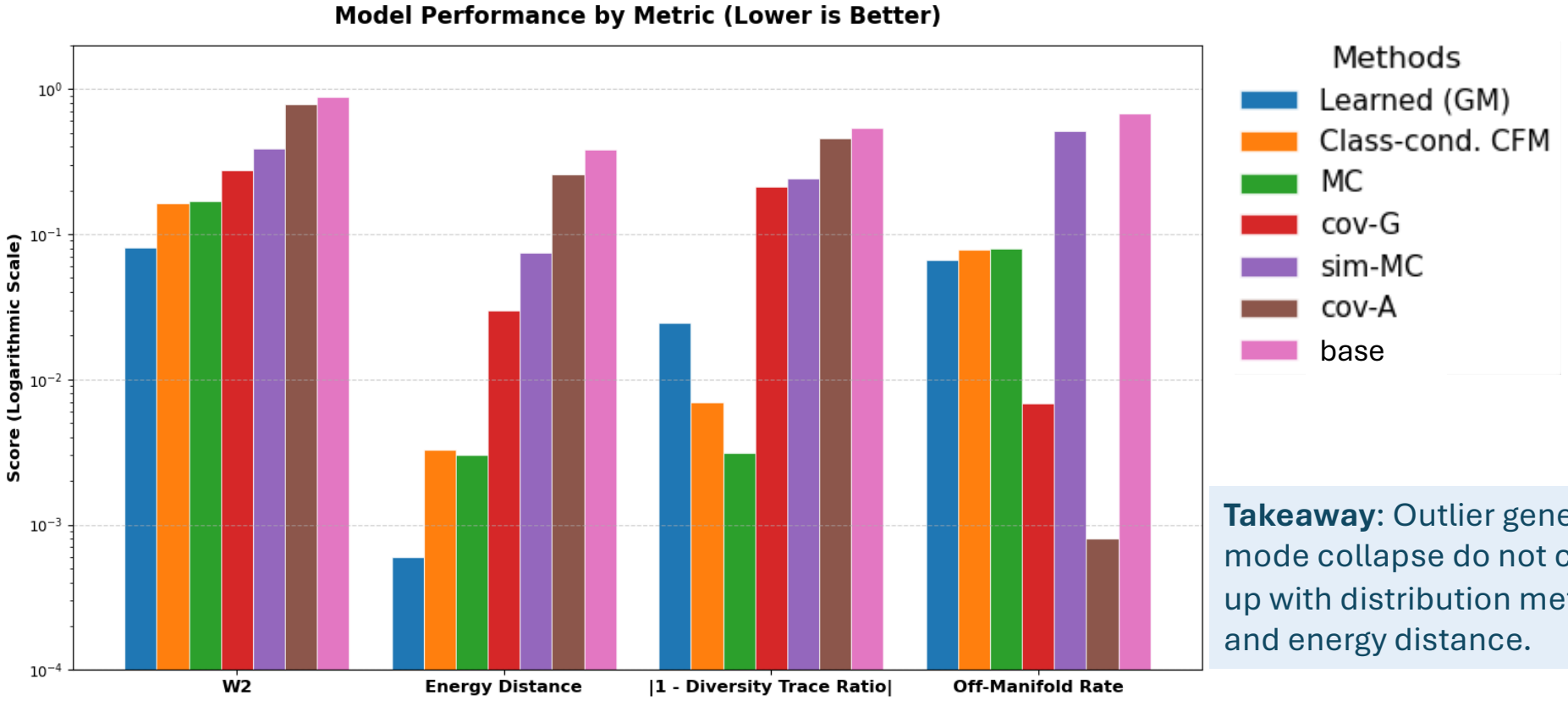
Results

Visual comparison of target control methods on toy data



Takeaway:
Learned (GM) performs best visually on toy data.
Guidance and class-conditional CFM (CCFM) perform similarly on toy examples.

Evaluating methods across complementary metrics



Takeaway: Outlier generation and mode collapse do not clearly show up with distribution metrics W2 and energy distance.

Minimum cost to transform one dist into another

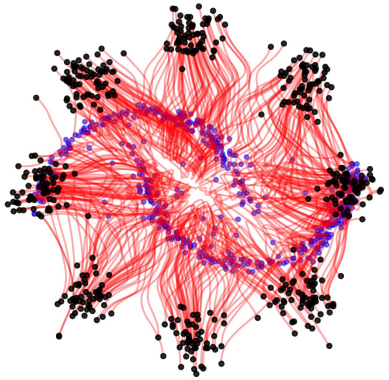
Mismatch in spatial overlap in terms of average distances

Change of variance between data and generation

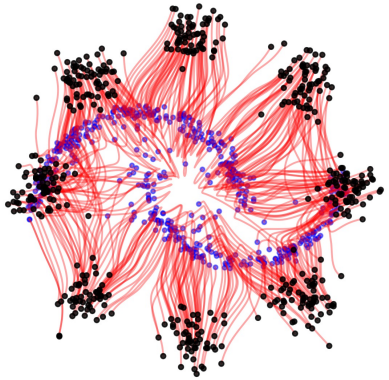
Frequency of (unrealistic) outliers

Overview of coupling strategies from independent to exact

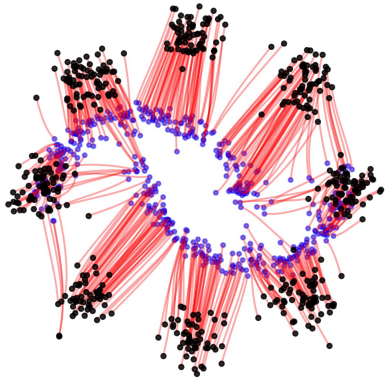
Independent-CFM

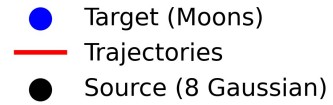


Entropic OT-CFM [4]



Exact OT-CFM [1]





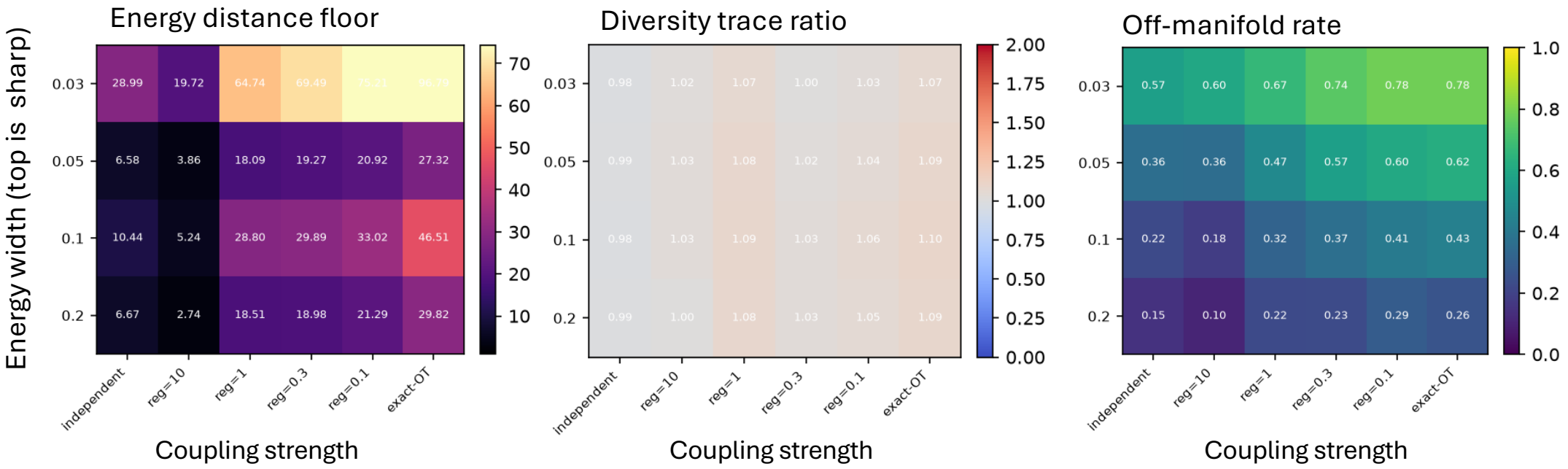
P = 1 *exactly* → *Coupling strength* → **P ≠ 1** *in general*

$$\pi(x_0, x_1) = p(x_0)p(x_1)$$
$$\pi'(x_0, x_1) = \frac{1}{Z}\pi(x_0, x_1)e^{-J(x_1)}$$

$$\mathcal{P} = \frac{\pi'(x_0|x_1)}{\pi(x_0|x_1)} \textit{ is the reverse coupling ratio}$$

Takeaway: energy guidance methods are only valid under the assumption P ≈ 1.
Single cell generation methods cover all three cases (CellOT exact OT maps, CellFlow entropic, OT-FM).

MC guidance performance depending on coupling strength and sharpness of energy function J



Takeaway: *MC's off-manifold rate climbs towards high coupling strength and sharp J.*

Conclusion

- Applied different guidance methods to conditional flow matching
- Learned Guidance (GM) performs slightly better than MC on synthetic data, but at the cost of additional training
- MC guidance with stronger OT coupling and sharp energy functions perform worse.

→ Careful testing is needed when applying guidance to flow matching models with non-independent coupling.

References

1. Tong et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, Transactions on Machine Learning Research, 2024
2. Lipman et al. Flow Matching for Generative Modeling, ICLR 2023
3. Feng et al. On the Guidance of Flow Matching, ICML 2025
4. Klein et al. GENOT: Entropic (Gromov) Wasserstein Flow Matching with Applications to Single-Cell Genomics, NeurIPS 2024

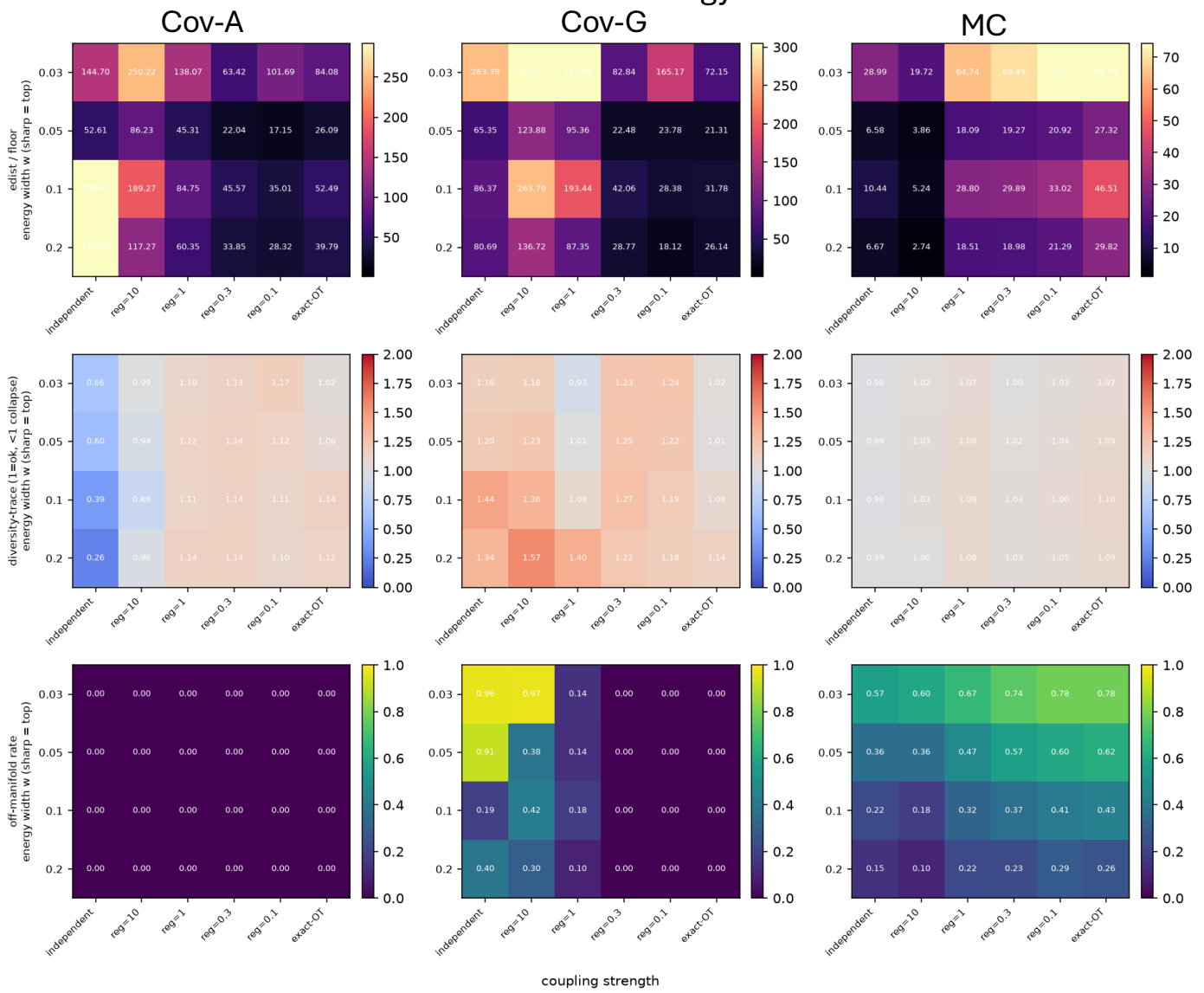
Questions?

Appendix

Appendix

Guidance-failure phase diagrams for different metrics

With uniform to 8-Gaussian distribution and with energy function for 3-Gaussians



Overview of different guidance methods

- The guidance vector field g_t should satisfy (Theorem 3.1of [3]):

$$g_t(x_t) = \int \left(\mathcal{P} \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) v_{t|z}(x_t|z) p(z|x_t) dz$$

Intractable integral due to $p(z|x_t)$

- Monte Carlo approximation of integral (sampling from prior instead of posterior):

$$\boldsymbol{g_t^{\mathrm{MC}}}(x_t) \triangleq \mathbb{E}_{x_1, x_0 \sim p(z)} \left[\left(\frac{e^{-J(x_1)}}{Z_t} - 1 \right) v_{t|z}(x_t|z) \frac{p_t(x_t|z)}{p_t(x_t)} \right]$$

- MC sim (approximate the posterior itself)

$$\boldsymbol{g_t^{\mathrm{MC\text{-}IS}}}(x_t) \triangleq \mathbb{E}_{x_1, x_0 \sim \tilde{p}(z)} \left[\frac{p(z)}{\tilde{p}(z)} \left(\frac{e^{-J(x_1)}}{Z_t} - 1 \right) \frac{p_t(x_t|z)}{p_t(x_t)} v_t(x_t|z) \right]$$

- Localize $p(x_1|x_t)$ via Taylor expansion of $Z_t \approx e^{-J(\hat{x}_1)}$

$$\boldsymbol{g_t^{\mathrm{local}}}(x_t) \triangleq -\mathbb{E}_{z \sim p(z|x_t)} \left[(x_1 - \hat{x}_1) v_{1|t}(x_t|z) \right] \nabla_{\hat{x}_1} J(\hat{x}_1)$$

- Covariance method; assume affine path $x_t = \alpha_t x_1 + \beta_t x_0 + \sigma_t \varepsilon$

$$\boldsymbol{g_t^{\mathrm{cov}}} \triangleq -\frac{\dot{\alpha}_t \beta_t - \dot{\beta}_t \alpha_t}{\beta_t} \Sigma_{1|t} \nabla_{\hat{x}_1} J(\hat{x}_1) \approx -\lambda_t^{\text{cov-A}} \nabla_{\hat{x}_1} J(\hat{x}_1)$$

with approx. $\Sigma' = \lambda_t I$

Overview of different guidance methods

Monte Carlo (MC)
- Monte Carlo estimation of intractable integral (due to $p(z|x_t)$) describing guidance vector field

Simulated Monte Carlo Gradient Guidance (sim-MC)
- approximates the intractable distribution with a gaussian
- Samples from this simpler shape to estimate guidance when the energy function lacks a clean math formula

Affine path covariance approximation (cov-A)
- Uses Taylor expansion to approximate guidance locally around the mean.
- Then approximates the covariance with hand-set schedule $\lambda_t I$

Affine path covariance approximation (cov-G)
- Assumes a standard Gaussian path to mathematically map flow matching back to classic diffusion guidance.
- Computes exact covariance using network gradients, offering strong performance at a higher computational cost.

Learned (GM)
- training-based method where a neural network is explicitly trained to become the guidance vector field
- Train multiple models: Z to estimate the normalizing constant, second model that directly learns the guidance vector field

Comparison of coupling methods

Feature	Exact OT	Entropic OT	Independent Coupling
Mapping Style	1-to-1 (Deterministic)	"Fuzzy" 1-to-Many (Probabilistic)	All-to-All (Fully Random)
Coupling Plan (π)	Highly sparse	Dense but localized	Fully dense ($\pi = p \times q$)
Vector Field Smoothness	Low (Sharp, rigid transitions)	Moderate to High (Tunable)	Very High (Averaged over all paths)
Training Variance	Low	Moderate (Balances structure and spread)	Extremely High
Computational Cost	High (Requires expensive solvers)	Low (Fast on GPUs via Sinkhorn)	Near Zero (Draw independent samples)
Primary Advantage	Straight, direct, non-crossing paths	Sweet spot between speed, variance, and smoothness	Trivial to implement and compute
Primary Drawback	Scales poorly to high dimensions	Requires tuning the ϵparameter	Neural network struggles to learn the massive web of paths

Additional slide